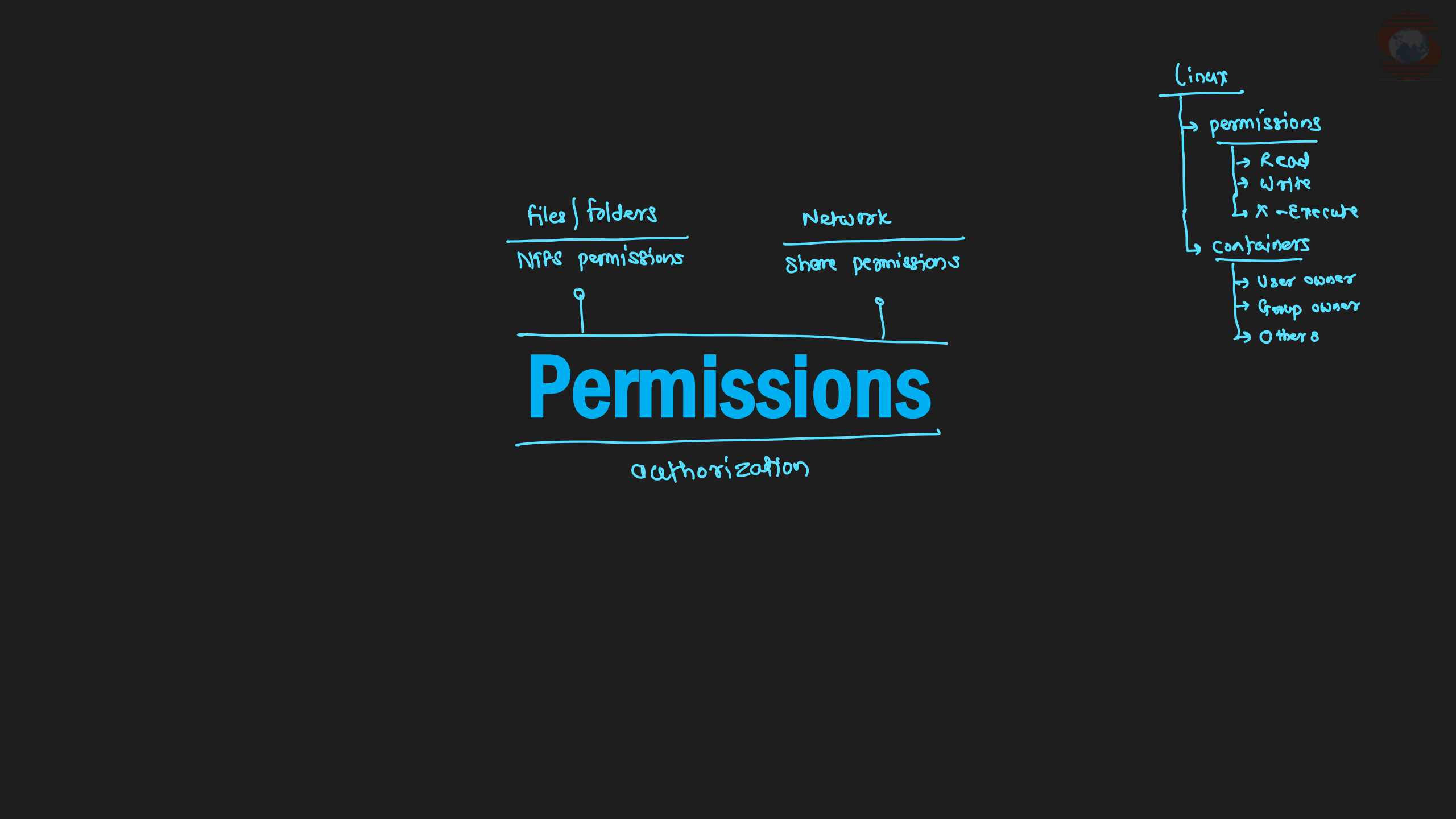

files / folders
NTFS permissions
Network
Share permissions
Permissions
authorization
Linux
permissions
Read
Write
X - Execute
containers
User owner
Group owner
Others

Introduction

final permissions = share + NTFS permission

- Windows Server 2022 uses **two independent permission systems** that combine to control access to shared resources:
  - server message block → protocol to share files over network → built-in feature of windows
- **Share Permissions (SMB-level):** Applied only when a folder is accessed over the network
- **NTFS Permissions (File System-level):** Applied to every file and folder, regardless of whether access is local or remote
- **The Golden Rule**
  - Deny
  - When a resource is accessed remotely, **the most restrictive permission wins**
  - If Share permissions grant "Read" but NTFS permissions grant "Modify," a network user gets **Read** access only
  - A user logged on locally to the server gets **Modify**, because Share permissions do not apply locally

Share Permissions (Network-Level Control)

- Share permissions are simpler and less granular than NTFS permissions
- They apply to the folder and all its contents as a single unit
- **Share Permission Levels**
  - Read: View file and subfolder names, navigate paths, view file data and attributes
  - Change: Everything Read allows, plus: Create files/folders, modify and append data, delete files/folders
  - Full Control: Everything Change allows, plus: Change NTFS permissions, take ownership
- **Best Practices**
  - **Keep Share permissions broad**
  - Set them to **Full Control** for the group that should access the share, and use **NTFS permissions to do the real fine-grained restriction**
  - This avoids confusing "double-restriction" scenarios where Share permissions block legitimate access that NTFS intended to allow
  - For most departmental shares
    - Set Share permission to **Change** for the department group and **Full Control** for Administrators
  - Only use Share-level restrictions when you need a hard network-only ceiling (e.g., a share that must be read-only for everyone over the network, even if local NTFS allows more)

NTFS Permissions (File System-Level Control)

- NTFS permissions are the primary tool for access control → applied on the child items
- They apply locally and remotely, and they support inheritance through folder hierarchies
- Basic NTFS Permission Levels
  - Basic Permission: Effectively Grants ***
  - Read: Read & Execute, List Folder Contents, Read. → Read file contents | get listing of folder
  - Read & Execute: Read, plus the ability to run executables. → execute a file
  - Write: Create files/folders, write data, append data. → update contents of file
  - Modify: Read & Execute, Write, and Delete. → delete a file from a folder
  - Full Control: Everything, plus Change Permissions and Take Ownership. → change of ownership
  - List Folder Contents: View folder contents only (applies only to folders). → view listing of folder

Critical Rules for Permission Evaluation

- **Rule 1: Cumulative Group Membership**
  - User's effective permissions are the **sum of all Allow permissions** from their user account and all groups they belong to
  - If User A has Read and is also a member of a group with Modify, User A gets **Modify**
- **Rule 2: Explicit Deny Wins**
  - An explicit **Deny** permission overrides any Allow permission, whether that Allow comes from a user account, a group, or inheritance
- **Rule 3: Explicit Allow Beats Inherited Deny**
  - An explicit **Allow** on a child object overrides an **inherited Deny** from a parent folder
  - This is a nuance that makes troubleshooting complex
- **Rule 4: Inheritance**
  - Child files and folders inherit permissions from their parent by default
  - Inherited permissions are shown with a faded/greyed checkbox in the GUI
  - You can break inheritance and copy the inherited permissions to become explicit, or remove them entirely

Access-Based Enumeration (ABE)

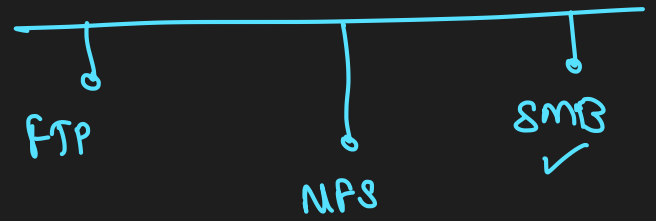


- ABE hides files and folders a user does not have **Read** access to, they simply do not appear in the directory listing
- **Scope:** Applies only to access via a share (SMB/DFS), not to local file system access
- **Effect:** It governs **visibility**, not access. If a user knows the full path, they can still attempt to open it—Share and NTFS permissions still govern whether they succeed
- **Requirement:** ABE works reliably only when NTFS permissions are set using **Active Directory groups**. Local groups (like local "Administrators") do not carry the per-user distinction ABE needs
- **Enablement:** It is configured per-share, usually when creating the SMB share in Server Manager or via Set-SmbShare -FolderEnumerationMode AccessBased

Quick Setup Checklist

- **Create the folder** on a data drive (not C:)
- **Share it** via Server Manager > File and Storage Services > Shares > New Share > SMB Quick
- **Enable ABE** on the share if users should only see what they can access
- **Set Share Permissions** to Full Control for the access group (let NTFS do the restriction)
- **Disable Inheritance** on the folder (if needed) and convert inherited permissions to explicit
- **Add Domain Local Groups** (e.g., Finance_Modify, Finance_Read) with the appropriate NTFS permissions
- **Verify with Effective Access** tab (Advanced Security Settings) to confirm a sample user's actual access

File Server



File Server

- The File Server role in Windows Server 2025 provides centralized storage and sharing of files across a network
- It encompasses file system management, the SMB (Server Message Block) protocol, security hardening, and resource management tools
- Windows Server 2025 introduces significant security enhancements to file services, particularly around SMB protocol hardening, making it the most secure Windows file server release to date

Core Components

- **File System**
  - Windows Server 2025 supports NTFS and ReFS
  - The choice depends on the intended use case—NTFS for general-purpose shares with advanced permissions, ReFS for large data sets and integrity-focused workloads
- **File Server Resource Manager (FSRM)**
  - FSRM provides quota management, file screening, and storage reporting.
  - It is an optional role service installed alongside the File Server role
- **SMB Protocol**
  - The primary network file-sharing protocol
  - Windows Server 2025 includes major security hardening and new connectivity options for SMB

Configuration Workflow

- **Step 1: Install the Role**
  - Use Server Manager → Add Roles and Features → **File and Storage Services** → **File Server**. Optionally add **File Server Resource Manager** for quota and screening capabilities
- **Step 2: Prepare Storage**
  - Create volumes/folders on data drives (not C:). Plan folder structure before sharing
- **Step 3: Configure SMB Sharing**
  - Use Server Manager or PowerShell (New-SmbShare)
  - Set Share Permissions broadly (Full Control or Change for access group)
  - Enable **Access-Based Enumeration (ABE)** if users should only see what they can access
- **Step 4: Apply NTFS Permissions**
  - Use the **AGDLP model**: Accounts → Global groups → Domain Local groups → Permissions. Assign permissions only to Domain Local groups, never individual users
- **Step 5: Configure FSRM (Optional)**
  - **Quotas**: Soft or hard limits per user/folder (e.g., 5 GB per user)
  - **File Screening**: Block media files or executables on data shares
  - **Notifications**: Email alerts on threshold breaches
- **Step 6: Enable VSS**
  - Configure shadow copies for previous versions and backup integration

DFS

- DFS Namespaces provide a unified namespace across multiple servers
- DFS Replication synchronizes data between servers
- **Key PowerShell commands**
  - New-DfsnFolder / New-DfsnFolderTarget — Create namespace folders and targets
  - New-DfsReplicationGroup — Create replication group
  - Add-DfsrConnection — Configure replication topology
  - Set-DfsrMembership — Assign content paths and primary member
- **Considerations**
  - Set staging quota (typically 16 GB per replicated folder)
  - Configure conflict and deleted quota
  - Schedule replication to avoid bandwidth saturation during business hours